

Yuyang He

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Education

Huazhong University of Science and Technology (HUST)

Sept 2024 – Jun 2028

B.S. in Materials Science and Engineering

Wuhan, China

- **GPA:** 3.37/5.0 **Ranking:** 54/98
- **Selected Coursework:** Quantum Mechanics(92), Solid State Physics(90), Fundamentals of Materials Science (90)
- **Research Interests:** Quantum information, Quantum many-body dynamics, Non-equilibrium quantum systems, Topological phases of matter

Research Experience

Studies of Non-Hermitian Systems

July 2026–Present

Research Assistant, HUST **Advisor:** Xiaoxiao Zhang

- **Literature Review:** Studied key reviews and representative papers in non-Hermitian physics, developing an understanding of core theoretical frameworks and analytical methods.
- **Numerical Replication:** Reproduced numerical results and corresponding plots from the non-Hermitian physics literature using Python, developing skills in numerical computation, algorithm implementation, and result analysis.
- **Academic Skills:** capable of reading synthesized English-language literature on non-Hermitian physics, prepared and shared reading notes on social media.

Low-Frequency Noise in Thermistors

Dec 2025 – Jun 2026

Research Assistant, HUST **Advisor:** Shaobo Qu

- **Research area:** Thermistor temperature calibration, lattice vibrations, and parameter estimation in low-frequency noise models, focusing on noise mechanisms and .
- **Framework:** Synthesized and presented literature and Performed preliminary parameter analysis of thermistor experimental data using deep learning.

Project Experience

Wafer-scale synthesis of two-dimensional semiconductors

June 2025 – June 2026

Core Member, HUST **Reward:** Silver Prize

- **Experimental Work:** Participated in MoS/WSe thin-film growth and SEM/XRD characterization, gaining hands-on experience in materials fabrication and optoelectronic performance analysis.

Honors & Awards

- Self-Improvement Scholarship of HUST Sept 2026
- Excellent Voluntary and Public Service Scholarship of HUST Sept 2026
- University Outstanding Communist Youth League Cadre Mar 2026
- Silver Prize, 14th "Qiusi Cup" Entrepreneurship Competition (*Top 10%*) Mar 2026
- Second Prize, 2025 SMSE Undergraduate Academic Conference Dec 2025

Leadership Experience

Director, Undergraduate Social Practice Center

Mar 2025 – Sept 2026

Coordinated teams and scheduling for large-scale volunteer events (e.g., the Wuhan Marathon).

Director, School Press & Media Publicity Center

Sept 2024 – Jun 2026

Directed new-media content planning and publishing; accumulated 140+ hours of signature Youth League activities.

Skills

Computational Tools: Python, MATLAB, Deep learning

Languages: English (CET-6: 514; CET-4: 540), Chinese (Native)